Accepted Manuscript

Synthesis, crystal structure, optical property and theoretical studies of a noncentrosymmetric telluromolybdate $CoTeMoO_6$

Chengguo Jin, Zhen Li

PII: S0925-8388(17)32149-7

DOI: 10.1016/j.jallcom.2017.06.150

Reference: JALCOM 42219

To appear in: *Journal of Alloys and Compounds*

Received Date: 12 April 2017

Revised Date: 11 June 2017

Accepted Date: 13 June 2017

Please cite this article as: C. Jin, Z. Li, Synthesis, crystal structure, optical property and theoretical studies of a noncentrosymmetric telluromolybdate $CoTeMoO_6$, *Journal of Alloys and Compounds* (2017), doi: 10.1016/j.jallcom.2017.06.150.

This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

ACCEPTED MANUSCRIPT

Synthesis, Crystal Structure, Optical Property and Theoretical Studies of a Noncentrosymmetric Telluromolybdate $CoTeMoO_6$

Chengguo Jin[1*, 2], Zhen Li[3]

[1]Key Laboratory of Computational Physics of Sichuan Province, Yibin University, Yibin 644000, China.

[2]School of Physics and Electronic Engineering, Yibin University, Yibin 644000, China.

[3]Faculty of Materials Science and Chemistry, China University of Geosciences, Wuhan 430074, China.

Corresponding Author: Chengguo Jin, E-mail: jcg.198406@163.com, Tel: +86 0831 353171.

Abstract: Polycrystalline $CoTeMoO_6$ has been synthesized, and $CoTeMoO_6$ crystals have been grown by spontaneous crystallization. Band structure, density of states, charge density, electronic structures and optical properties of $CoTeMoO_6$ crystal were calculated by using a first-principles method for the first time. Optical and nonlinear optical properties of $CoTeMoO_6$ crystal were measured. Structure and charge density analyses indicate that cleavage maybe exist in $CoTeMoO_6$ crystal. The experiment results of optical properties show that a strong absorption band exhibits at 458~692 nm in the visible region. The absorption band plays an important role in the coloration of $CoTeMoO_6$ crystal. The SHG responses of $CoTeMoO_6$ with a 1.06μm laser is much weaker than that of KDP, which can be ascribed to the absorption band caused by spin-allowed *d-d* transitions of Co^{2+} ions. Thus $CoTeMoO_6$ crystal is not suited to be used as nonlinear optical material in the Vis-NIR region.

Keywords: A. inorganic materials; B. crystal growth; C. optical properties; D. nonlinear optics; D. computer simulations.

1 Introduction

A$TeMoO_6$ (*A*=Mn, Mg, Zn, Cd, Co) is a remarkable family of noncentrosymmetric (NCS) telluromolybdate. It is proved that $MnTeMoO_6$, $MgTeMoO_6$, $ZnTeMoO_6$, $CdTeMoO_6$ exhibit excellence nonlinear optical (NLO) properties [1-4]. $CoTeMoO_6$ crystal has almost the same structure as $MgTeMoO_6$ and $ZnTeMoO_6$, and contains d^0 transition metal cations (Mo^{6+}) and cations with nonbonded electron pairs (Te^{4+}). It is reported that the combination of above two types of cations in the same compound can lead to compounds with enhanced second harmonic generation (SHG) properties [5], so $CoTeMoO_6$ crystal is a candidate for NLO material. Polycrystalline $CoTeMoO_6$ was firstly synthesized by Kozlowski and Sloczynski [6] in the 1970s, and was focused on its catalytic and magnetic properties [7-8]. $CoTeMoO_6$ crystal was firstly grown by Mączka *et al* [9] with spontaneous crystallization about 40 years later, and lattice dynamics and high-pressure Raman scattering of $CoTeMoO_6$ crystal were investigated [10]. However, it is regrettable that $CoTeMoO_6$ crystal cannot exhibit second harmonic generation (SHG) response, which reported by Mączka *et al* [9]. Therefore, it is necessary to study the absence reason of SHG in $CoTeMoO_6$, which is beneficial to the application of this crystal. In addition, the theoretical studies of electronic structures can be better to investigate the crystal

structure and the absence reason of SHG.

In this work, we present a detailed investigation on the band structure, density of states, charge density and electronic structures of $CoTeMoO_6$ for the first time. Both experimental and theoretical studies were performed to investigate the optical properties of $CoTeMoO_6$. The nonlinear optical behavior in $CoTeMoO_6$ crystal was reported, furthermore, the absence reason of SHG was discussed.

2 Experimental and computational details

2.1 Synthesis and crystal growth

Polycrystalline $CoTeMoO_6$ has been synthesized with CoO, TeO_2 and MoO_3 by traditional solid-state reaction techniques. These raw materials were weighed in stoichiometric ratios, and ground intimately in an agate mortar. The mixtures were sintered three times at 500℃ for 20 h with intermediate grinding. The single phase powder exhibits a violet color. The resultant (Fig.1a) is in good agreement with the JCPDS card 32-0320 (Fig.1c).

$CoTeMoO_6$ crystals for single crystal X-ray diffraction measurement have been grown from high temperature solution by means of spontaneous crystallization. The TeO_2-MoO_3 mixture was used as self-flux system. $CoTeMoO_6$, TeO_2 and MoO_3 in a molar ratio of 1:3:3 were mixed together and contained in a platinum crucible. The mixture was heated up to 720℃ and held for 2 days to melt the powders to a homogeneous liquid solution. During the growth process, the temperature decreasing rate is 0.2~0.5 °C per day.

2.2 Characterizations

The structure of $CoTeMoO_6$ was determined by standard crystallographic methods. A violet crystal with dimensions of 0.20×0.20×0.04 mm^3 was chosen for structure determination. The structure data were collected on a Bruker SMART APEX-II diffractometer equipped with a CCD area detector using graphite-monochromated Mo-Kα radiation (λ=0.71073 Å) at 100(2) K. The structure was solved by direct methods and refined by full matrix least-squares methods on F^2 by SHELX-97 [11]. Crystallographic data and structural refinements are listed in Table 1. Final refined atomic coordinates and isotropic displacement parameters are summarized in Table S1. Selected bond distances (Å) and angles (deg) are given in Table S2, and anisotropic displacement parameters are listed in Table S3. Additional information in the form of CIF is available in the supplementary material.

Powder X-ray diffraction data were collected using a Rigaku MiniFlex 600 X-ray diffractometer (Cu Kα radiation; λ=1.54056 Å) over the 2θ range 10~65° at room temperature.

UV-vis spectra were obtained from a SHIMADZU UV-2550 spectrometer equipped with an integrating sphere over the spectral in the range 240~850 nm at room temperature. $BaSO_4$ was used as a reference material, on which $CoTeMoO_6$ powders were coated. The absorption spectra were calculated by the reflectance spectra using the Kubelka-Munk function [12].

Powder second harmonic generation measurements were performed on the modified method of Kurtz and Perry [13]. A Q-switched Nd:YAG laser (λ=1.06 μm) was used as the fundamental wave to determine the SHG efficiency. Polycrystalline $CoTeMoO_6$ were ground and sieved into 150~200 μm. Sieved KDP powder (150~200 μm) was used for comparing SHG intensities.

2.3 Computational details

Single-crystal structural data of $CoTeMoO_6$ was used for the theoretical calculations. All

calculations were performed with the commercial software CASTEP code in Materials Studio of Accelrys Inc [14-15] by using the first-principles calculations based on density functional theory (DFT) [16-17]. For the exchange-correlation functional, the ultrasoft pseudo-potential [18] and the local density approximation (LDA) parameterized with the Ceperley-Alder-Perdew-Zunger (CAPZ) [19-20] functional were adopted. The $3d^7 4s^2$, $5s^2 5p^4$, $4d^5 5s^1$ and $2s^2 2p^4$ electrons were taken as the valence electrons of Co, Te, Mo and O atoms, respectively. In order to consider the localized states in the transition elements Co and Mo, the DFT+*U* [21] method with an additional on-site *d* orbital dependent correlation Hubbard *U* was adopted to calculate the electronic structure in $CoTeMoO_6$ as well. Kinetic energy cutoff 380 eV and Monkhorst-Pack [22] *k*-point meshes with a density of 3×3×2 points in the Brillouin zone were chosen. The other calculation parameters and convergent criteria were the default values of the CASTEP code.

3 Results and discussion

3.1 Crystal structure

$CoTeMoO_6$ crystallizes in the tetragonal space group $P2_12_12$ (No.18) with cell parameters a = 5.269(3) Å, b = 5.269(3) Å, c = 9.014(11) Å and Z =2 (show in Table 1). There obtained parameters and bond lengths at 100 K in this work (Tables S1~S3) are very similar to those results measured at 298 K and 150 K [9]. X-ray diffraction patterns of polycrystalline $CoTeMoO_6$ (Fig.1a) are in good agreement with the calculated patterns derived from the single crystal data (Fig.1b), indicating that the results in the present study are exact.

As shown in Fig.2, there are one Co atom, one Te atom and one Mo atom in the asymmetric unit. The Co atom is coordinated by six oxygen atoms to form an octahedral geometry in Fig.2a. The Co-O bond distances range from 2.078(5) to 2.210(6) Å. The Te atom is three-coordinate to display see-saw geometry (Fig.2b) with the Te-O bond distances ranging from 1.894(6) to 2.088(6) Å. The Mo atom is coordinated by four oxygen atoms between 1.704(6) and 1.859(6) Å to form a distorted tetrahedral coordination (Fig.2c).

Similar to $MgTeMoO_6$ [2] and $MnTeMoO_6$ [23], $CoTeMoO_6$ exhibits a neutral layered structure composed of CoO_6 octahedra linked to asymmetric TeO_4 polyhedra and MoO_4 tetrahedra (Fig.2e). These layers are further held together to form three-dimensional network and a *gap* parallels to (002) is located between the neutral layers in $CoTeMoO_6$.

3.2 Band structure, DOS and Charge density

The band structure of $CoTeMoO_6$ (Fig.3) shows that the top of the valence bands (VBs) and the bottom of the conduction bands (CBs) are at the different point. The lowest energy of the CBs (1.931 eV) is at the Q point, whereas the highest energy (0.0 eV) of the VBs is located at the Z point. Therefore, $CoTeMoO_6$ shows a semiconducting character with an indirect band gap of 1.93 eV.

The total and partial density of states (PDOS) is shown in Fig.4. The VBs between -20 and -15 eV is mostly formed by O 2*s*, mixings with small Te 5*s* and Mo 4*d* states. The VBs lying about between -12 to -9 eV is derived from Te 5*s* and O 2*s* 2*p*. The VBs from -8 eV up to the Fermi energy are mainly composed of O 2*p* and Co 3*d*, with small mixings of Te 5*p* states. The CBs above the Fermi level (from 2 to 6 eV) are derived from O 2*p*, Co 3*d* and Mo 4*d*. Accordingly, the optical absorption of $CoTeMoO_6$ is mainly ascribed to the charge transitions from O 2*p* to Co 3*d*, Te 5*p* and Mo 4*d* states, and from VBs to CBs in Co 3*d*.

The charge density can give an indication of the likelihood of formation of bonding and non-bonding electron pairs [24-26]. The total charge density is presented in Fig.5. There is almost no electron accumulation between the oxygen atoms with the two proximate neutral layers (Fig.5a), which demonstrates that no chemical bonds are formed between them.

In order to give an intuitional insight into the electronic interactions between the two proximate neutral layers, the total charge density of $CoTeMoO_6$ for a selected (002) and (110) plane are shown in Fig.5b and Fig.5c respectively. These four oxygen atoms in the (002) plane are insular, and Mo atom cannot connect with the oxygen atom in the different layer in the (110) plane. It indicates that there is no bonding action between the two proximate neutral layers. Thus, these layers can only be held together by *van der Waals* forces to form a weak three-dimensional network. It can infer that a natural cleavage plane maybe exist in $CoTeMoO_6$ crystal which is similar to $MnTeMoO_6$ crystal [23].

3.3 Optical property

In order to understand the reason of SHG absence, both experimental and theoretical studies were employed to investigate the optical properties of $CoTeMoO_6$.

The UV-vis diffuse reflectance spectra of $CoTeMoO_6$ are shown in Fig.6. It is worth noting that there is an absorption band locate approximately at 458~692 nm (2.71~1.80 eV) in the visible region, and the absorption peak locate at 530~610 nm. According to Fig.4, the absorption band (2.71~1.80 eV) are mainly caused by the electron transition from VBs to CBs in Co 3*d* orbital, indicating that the absorption band in $CoTeMoO_6$ would be attributed to spin-allowed *d-d* transitions of Co^{2+} ($3d^7$ electronic configuration) in octahedral crystal field of oxygen ligands [27-30]. As we known, when an absorption band removes a certain color from the transmitted beam, the eye collects the remaining colors and produces the complementary color to the one removed [31]. Thus when green to orange light at 530~610 nm is absorbed by Co^{2+} in $CoTeMoO_6$, the crystal appears a violet complementary color. Thus, it can be inferred that the absorption peak caused by spin-allowed *d-d* transitions of Co^{2+} ions plays an important role in the coloration of $CoTeMoO_6$ crystal.

$CoTeMoO_6$ is an indirect band gap semiconductor, so the band gap, the absorption and the wave frequency all obey the following equation [32]: $(\alpha h\upsilon)^{1/2}=A(h\upsilon-E_g)$, near the cut-off of the optical transmission. The $(\alpha h\nu)^{1/2}$-$(h\nu)$ curve is drawn on the upper right insert of Fig. 6 based on the literature [32]. The band gap energy of $CoTeMoO_6$ is determined to be approximately 2.29 eV by fixing the tangent line of the curve and the $(h\nu)$ axis, which is comparable with the calculated value (1.93 eV). It is well known that the band-gap calculated by DFT is usually smaller than the experimental data due to the discontinuity of exchange-correlation energy [33].

The absorption spectra are calculated with a scissor operator of 0.36 eV, and shown in Fig.7. The curve also exhibits a strong absorption band, however, the absorption band in the calculated result is at 105~400 nm, which is different with that in the experimental result. It is due to the limitation of the DFT method that sometimes underestimates the band gap in semiconductors and insulators [34].

3.4 SHG capability

In order to investigate the nonlinear optical (NLO) properties of $CoTeMoO_6$, the powder SHG measurements were performed with a 1.06μm Nd:YAG laser. Repeated testing, however, it has

been found that the SHG response of $CoTeMoO_6$ is much weaker than that of KDP, and the value is close to zero comparing with KDP.

As we know, both Mo^{6+} and Te^{4+} cations are susceptible to SOJT distortions [5], and the SOJT distortion leads to strong NLO properties in some noncentrosymmetric telluromolybdates, such as: $MgTeMoO_6$ [2], $BaTeMo_2O_9$ [35], $Cs_2TeMo_3O_{12}$ [36]. However, it is strange that the SHG capability is disappeared in $CoTeMoO_6$. According to Fig.6, a strong absorption band at 458~692 nm was exhibited, which would leads to the absorbing and disappearance of the output efficiency of 532 nm frequency-doubling green lasers. Thus, the SHG response of 1.06μm would quickly vanish. It is proved that the SHG absence is mostly ascribed to the strong absorbing at 458~692 nm. Finally, it is ascribed to spin-allowed *d-d* transitions of Co^{2+}. It demonstrates that the application of $CoTeMoO_6$ has important limitation in the frequency doubling with 1.06μm laser. Therefore, $CoTeMoO_6$ crystal is not suited to be used as NLO material in the Vis-NIR region.

4 Conclusions

Polycrystalline $CoTeMoO_6$ has been synthesized and $CoTeMoO_6$ crystals have been grown by spontaneous crystallization method. Band structure, density of states, charge density, electronic structures and optical properties of $CoTeMoO_6$ crystal have been studied. Optical and nonlinear optical properties of $CoTeMoO_6$ crystal are investigated. Structure and charge density analysis results indicate that a natural cleavage plane maybe exist in $CoTeMoO_6$ crystal, which is similar to $MnTeMoO_6$. The experiment results of optical properties show a strong absorption band at 458~692 nm in the visible region. The absorption band plays an important role in the coloration of $CoTeMoO_6$ crystal. The SHG response of $CoTeMoO_6$ with a 1.06μm laser is much weaker than that of KDP, which is ascribed to the absorption band caused by spin-allowed *d-d* transitions of Co^{2+} ($3d^7$ electronic configuration) in octahedral crystal field of oxygen ligands. Thus, the application of $CoTeMoO_6$ as nonlinear optical material would be limited in the Vis-NIR region.

Acknowledgements

This work was supported by the Scientific Research Key Project of Yibin University [No.2015QD11] and Major Projects of Yibin City of China.

References

[1] C. G. Jin, Z. Li, L. X. Huang, M. Z. He, Top-seeded solution growth and characterization of a novel nonlinear optical crystal $MnTeMoO_6$, J. Cryst. Growth 369(2013)43-46.

[2] J. J. Zhang, Z. H. Zhang, Y. X. Sun, C. Q. Zhang, S. J. Zhang, Y. Liu, X. T. Tao, $MgTeMoO_6$: a neutral layered material showing strong second-harmonic generation, J. Mater. Chem. 22(2012): 9921-9927.

[3] S. G. Zhao, J. H. Luo, P. Zhou, S. Q. Zhang, Z. H. Sun, M. C. Hong, $ZnTeMoO_6$: a strong second-harmonic generation material originating from three types of asymmetric building units, RSC Adv. 3(2013)14000-14006.

[4] S. G. Zhao, X. X. Jiang, R. He, S. Q. Zhang, Z. H. Sun, J. H. Luo, Z. S. Lin, M. C. Hong, A combination of multiple chromophores enhances second-harmonic generation in a nonpolar noncentrosymmetric oxide: $CdTeMoO_6$, J. Mater. Chem. C 1(2013) 2906-2912.

[5] P. S. Halasyamani, K. R. Poeppelmeier, Non-centrosymmetric oxides, Chem. Mater. 10 (1998) 2753-2769.

[6] R. Kozlowski, J. Sloczynski, $CoTeMoO_6$, $Co_4TeMo_3O_{16}$: two new cobalt molybdotellurates, J. Solid State Chem. 18(1976)51-55.

[7] P. Forzatti, F. Trifiro, P. L. Villa, $CdTeMoO_6$, $CoTeMoO_6$, $MnTeMoO_6$, and $ZnTeMoO_6$: a new class of selective catalysts for allylic oxidation of butene and propylene, J. Catal., 55(1978) 52-57.

[8] Y. Doi, R. Suzuki, Y. Hinatsu, K. Ohoyama, Crystal structures and magnetic properties of two-dimensional antiferromagnets $Co_{1-x}Zn_xTeMoO_6$, J. Solid State Chem.182(2009) 3232-3237.

[9] M. Mączka, K. Hermanowicz, A. Majchrowski, Ł. Kroenke, A. Pietraszko, M. Ptak, Growth and characterization of nonlinear optical telluromolybdate $CoTeMoO_6$ single crystals, J. Solid State Chem. 220(2014)142-148.

[10] K. Szymborska-Małek, M. Mączka, T. A. da Silva, W. Paraguassu, A. Majchrowski, Lattice dynamics and high-pressure Raman scattering studies of $CoTeMoO_6$ crystal, Vib. Spectrosc. 84(2016)153-158.

[11] G. M. Sheldrick, *SHELXTL, Crystallographic Software Package, SHELXTL, Version 5.1*, Bruker-AXS, Madison, WI, 1998.

[12] H. G. Hecht, W. M. Wendlandt, *Reflectance spectroscopy*, Interscience, New York, 1966.

[13] S. K. Kurtz, T. T. Perry, A powder technique for the evaluation of nonlinear optical materials, J. Appl. Phys. 39(1968)3798-3813.

[14] S. J. Clark, M. D. Segall, C. J. Pickard, P. J. Hasnip, M. J. Probert, K. Refson, M. C. Payne, First principles methods using CASTEP, Z. Kristallogr. 220 (2005) 567-570.

[15] D. Srivastva, M. Menon, K. Cho, Nanoplasticity of single-wall carbon nanotubes under uniaxial compression, Phys. Rev. Lett. 83 (1999) 2973-2976.

[16] P. Hohenberg, W. Kohn, Inhomogeneous electron gas, Phys. Rev. B 136 (1964) 864-871.

[17] W. Kohn, L. J. Sham, Self-consistent equations including exchange and correlation effects, Phys. Rev. A 140 (1965) 1133-1138.

[18] D. Vanderbilt, Soft self-consistent pseudopotentials in a generalized eigenvalue problem, Phys. Rev. B 41 (1990) 7892-7895.

[19] J. P. Perdew, A. Zunger, Self-interaction correction to density-functional approx-imations for many-electron systems, Phys. Rev. B 23 (1981) 5048-5079.

[20] D. M. Ceperley, B. J. Alder, Ground state of the electron gas by a stochasticmethod, Phys. Rev. Lett. 45 (1980) 566-569.

[21] V. I. Anisimov, J. Zaanen, O. K. Andersen, Band theory and mott insulators: hubbard U instead of stoner I, Phys. Rev. B 44 (1991) 943-954.

[22] H. J. Monkhorst, J. D. Pack, Special points for Brillouin-zone integrations, Phys. Rev. B 13 (1976) 5188-5192.

[23] C. Jin, L. Huang, J. Yang, M. Wan, Z. Li, Q. Cao, D. Huang, J. Shao, F. Wang, Relationship between structure and cleavage behavior in the nonlinear optical crystal $MnTeMoO_6$, J. Cryst. Growth 419(2015) 25-30.

[24] T. B. Terriberry, D. F. Cox, D. A. Bowman, A tool for the interactive 3D visualization of electronic structure in molecules and solids, Comput. Chem. 26(2002) 313-319.

[25] D. Pathak, T. Wagner, T. Adhikari, J. M. Nunzi, $AgInSe_2$. PCBM. P3HT inorganic organic blends for hybrid bulk heterojunction photovoltaics, Synthetic Met. 200(2015):102-108.

[26] D. Pathak, R. K. Bedi, D. Kaur, R. Kumar, Fabrication of densely distributed silver indium selenide nanorods by using Ag^+ ion irradiation, J. Korean Phys. Soc. 57(2010): 474-479.

[27] M. Wildner, Polarized electronic absorption spectra of Co^{2+} ions in the kieserite-type compounds $CoSO_4 \cdot H_2O$ and $CoSeO_4 \cdot H_2O$, Phys. Chem. Miner. 23(1996)489-496.

[28] D. Pathak, R. K. Bedi, D. Kaur, Effect of substrate temperature on the structural, optical, and electrical properties of silver-indium-selenide films prepared by using laser ablation, J. Korean Phys. Soc. 56(2010):836-841.

[29] M. J. L. Percival, E. Salje, Optical absorption spectroscopy of the $P2_13$-$P2_12_12_1$ transformation in $K_2Co_2(SO_4)_3$ langbeinite, Phys. Chem. Miner. 16(1989)563-568.

[30] D. Pathak, T. Wagner, T. Adhikari, J. M. Nunzi, Photovoltaic performance of $AgInSe_2$-conjugated polymer hybrid system bulk heterojunction solar cells, Synthetic Met. 199(2015)87-92.

[31] R. Newnham, *Properties of Materials: Anisotropy, Symmetry, Structure*, Oxford, Oxford University Press, 2005.

[32] M. A. Butler, Photoelectrolysis and physical properties of the semiconducting electrode WO_2, J. Appl. Phys. 48(1977)1914.

[33] R. Cong, T. Yang, F. Liao, F. Liao, Y. Wang, Z. Lin, J. Lin, Experimental and theoretical studies of second harmonic generation for $Bi_2O_2[NO_3(OH)]$, Mater. Res. Bull. 47(2012) 2573-2578.

[34] Y. Zhou, C. L. Hu, T. Hu, F. Kong, J. G. Mao, Explorations of new second-order NLO materials in the Ag^{I}-Mo^{VI}/W^{VI}-Te^{IV}-O systems, Dalton T. 29(2009)5747-5754.

[35] W. G. Zhang, X. T. Tao, C. Q. Zhang, H. J. Zhang, M. H. Jiang, Structure and thermal properties of the nonlinear optical crystal $BaTeMo_2O_9$, Cryst. Growth Des. 9(2009)2633-2636.

[36] J. J. Zhang, X. T. Tao, Y. X. Sun, Z. H. Zhang, C. Q. Zhang, Z. L. Gao, H. B. Xia, S. Q. Xia, Top-seeded solution growth, morphology, and properties of a polar crystal $Cs_2TeMo_3O_{12}$, Cryst. Growth Des. 11(2011)1863-1868.

ACCEPTED MANUSCRIPT

Table Captions

Table 1 Crystallographic data and structural refinements for $CoTeMoO_6$

Table S1 Atomic coordinates and equivalent isotropic displacement parameters ($Å^2$) for $CoTeMoO_6$ determined using the single-crystal XRD

Table S2 Selected bond lengths (Å) and angles (deg.) for $CoTeMoO_6$[a]

Table S3 Anisotropic displacement parameters ($Å^2 \times 10^3$) for $CoTeMoO_6$

Figure Captions

Fig.1 (a) Experimental, (b) calculated and (c) standard powder X-ray diffraction patterns of $CoTeMoO_6$

Fig.2 (a) CoO_6 octahedra, (b) TeO_4 polyhedra ,(c) MoO_4 polyhedral, (d) Unit cell diagram of $CoTeMoO_6$ with the ball-and-stick representation, and (e) three-dimensional network of the $CoTeMoO_6$ crystal structure.

Fig.3 Band structure of $CoTeMoO_6$ (bands are shown only between -5 and 5 eV for clarity).

Fig.4 Total and partial density of states of $CoTeMoO_6$.

Fig.5 (a) Charge density distributions of $CoTeMoO_6$ crystal structure; (b) total charge density of $CoTeMoO_6$ for a selected (002) plane; (c) total charge density of $CoTeMoO_6$ for a selected (110) plane.

Fig.6 UV-vis diffuse reflectance spectroscopy plots for $CoTeMoO_6$. The top right inset shows the plots of $(\alpha h\nu)^{1/2}$ vs. energy ($h\nu$) for the band gap energies. Note that the band gap is 2.29 eV for $CoTeMoO_6$.

Fig.7 Calculated result of absorption spectra for $CoTeMoO_6$.

ACCEPTED MANUSCRIPT

Table 1 Crystallographic data and structural refinements for $CoTeMoO_6$ [a]

Formula	$CoTeMoO_6$
Formula weight (g/mol)	378.47
Temperature (K)	100(2)
Radiation, wavelength (Å)	Mo Kα, 0.71073
Crystal system	Orthorhombic
Space group	$P2_12_12$ (No.18)
Unit cell dimensions (Å)	a = 5.059(3)
	b = 5.243(3)
	c = 8.812(6)
Unit cell volume (Å3)	233.7(3)
Z	2
D_c (g/cm^3)	5.377
Absorption coefficient (mm^{-1})	12.291
F(000)	339
Crystal size (mm^3)	0.20 × 0.20 × 0.04
Index ranges	$-7 \le h \le 7$, $-7 \le k \le 7$, $-12 \le l \le 11$
Reflections collected	2138
Independent reflections	740 (R_{int} = 0.0485)
Absorption correction	None
Refinement method	Full-matrix least-squares on F^2
Data / restraints / parameters	740 / 0 / 43
Goodness-of-fit on F^2	1.120
R indices [$I>2\sigma(I)$][a]	R_1 = 0.0396, wR_2 = 0. 0.1245
R indices (all data)[a]	R_1 = 0.0399, wR_2 = 0.1249
Absolute structure parameter	0.09(9)
Largest diff. peak and hole (e. Å^{-3})	2.769 and -2.919

[a] $R_1 = \Sigma\left\|F_o\right| - \left|F_c\right\| / \Sigma\left|F_o\right|$, $wR_2 = \{\sum w[(F_0)^2 - (F_c)^2]^2 / \sum w[(F_o)^2]^2\}^{1/2}$

ACCEPTED MANUSCRIPT

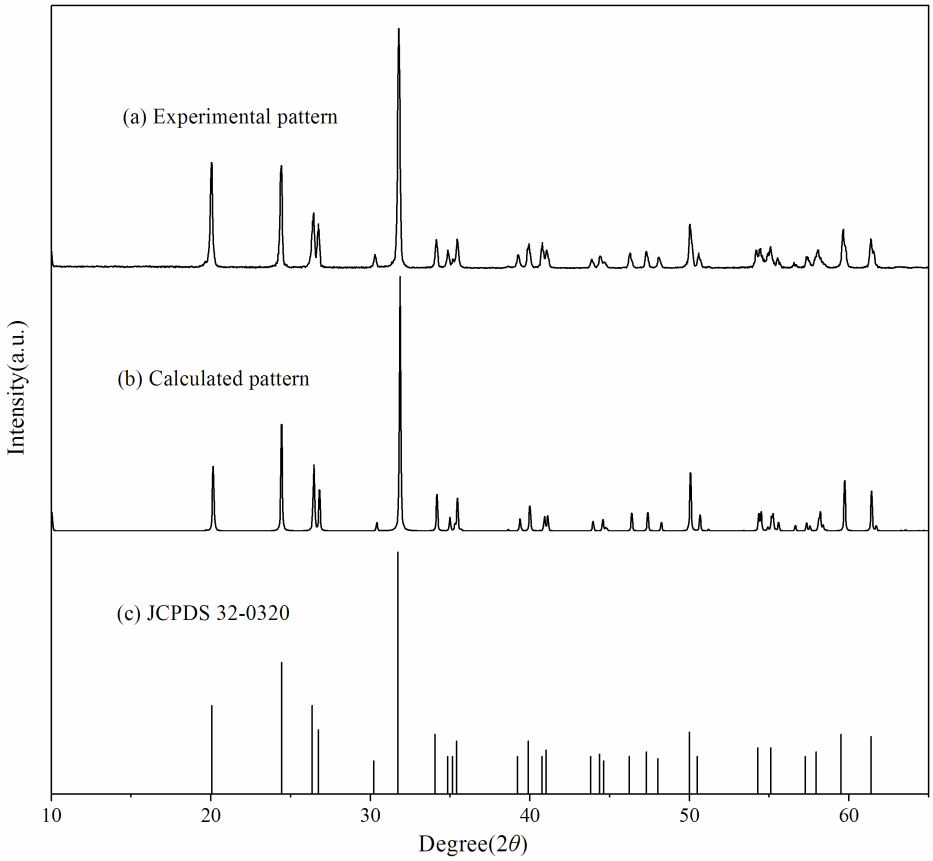

Fig.1 (a) Experimental, (b) calculated and (c) standard powder X-ray diffraction patterns of $CoTeMoO_6$

ACCEPTED

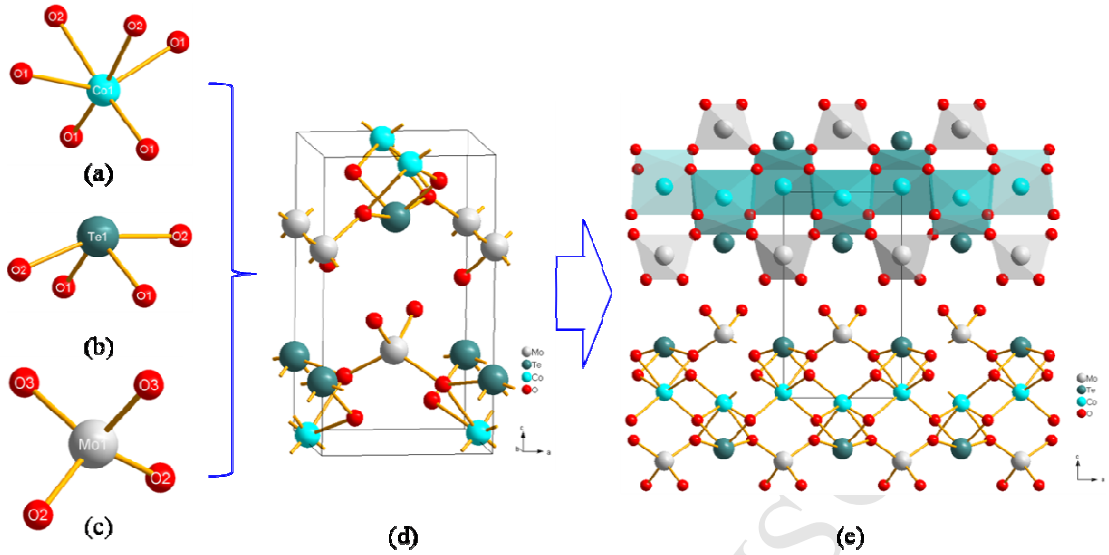

Fig.2 (a) CoO_6 octahedra, (b) TeO_4 polyhedra ,(c) MoO_4 polyhedral, (d) Unit cell diagram of $CoTeMoO_6$ with the ball-and-stick representation, and (e) three-dimensional network of the $CoTeMoO_6$ crystal structure.

ACCEPTED MANUSCRIPT

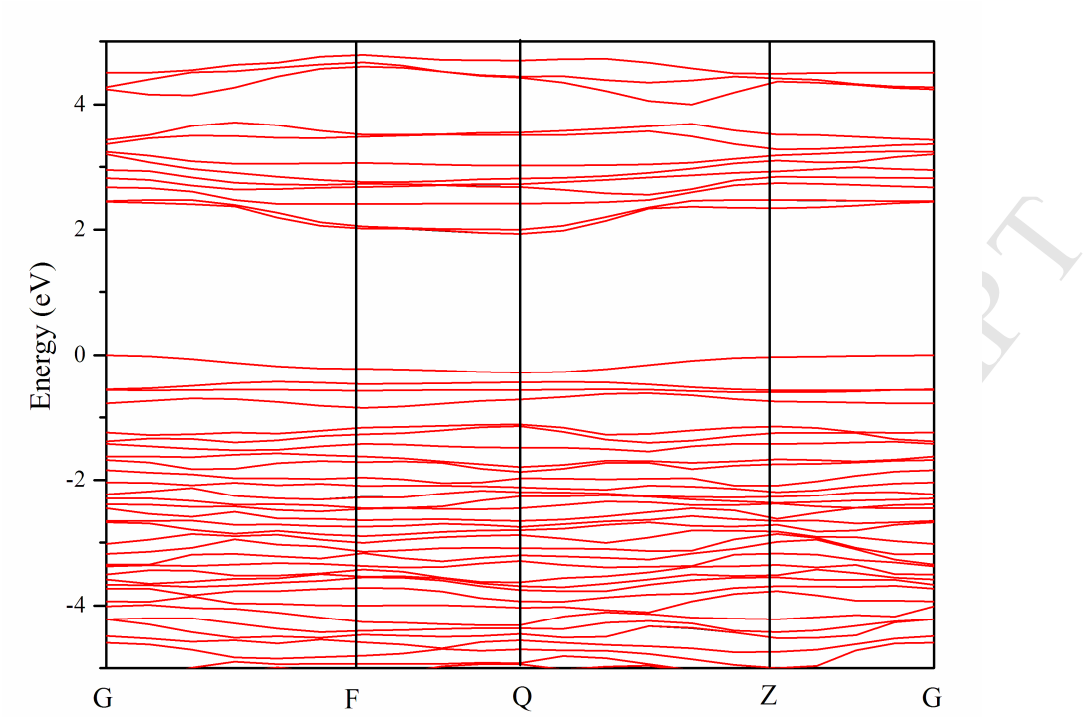

Fig.3 Band structure of $CoTeMoO_6$ (bands are shown only between -5 and 5 eV for clarity).

ACCEPTED MANUSCRIPT

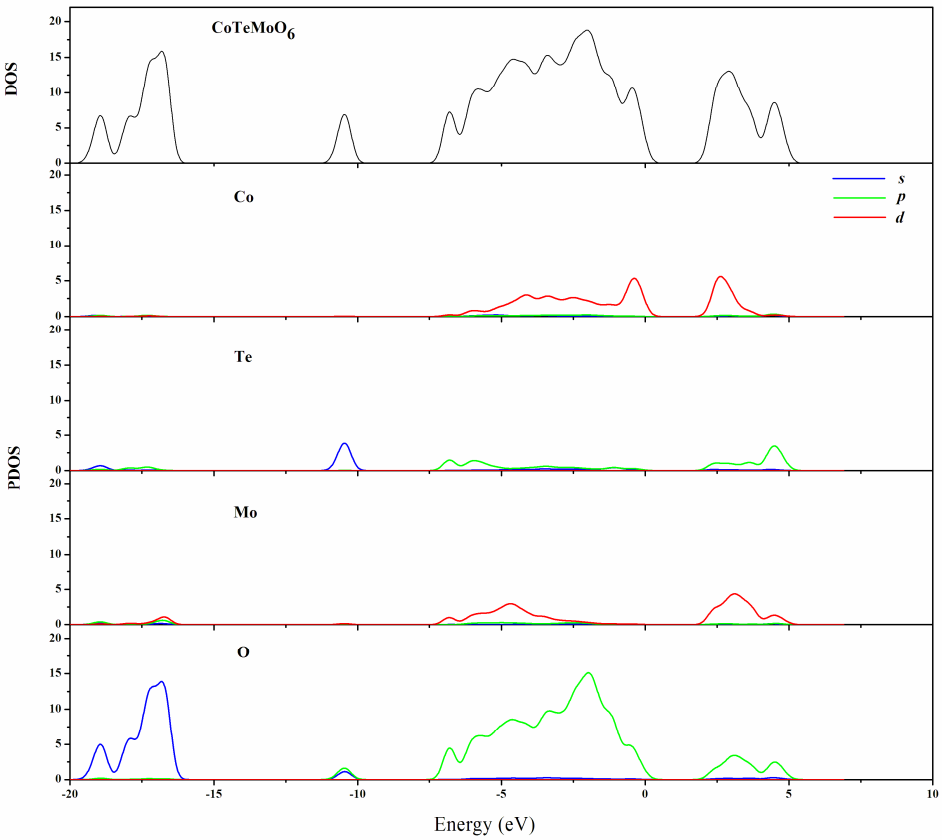

Fig.4 Total and partial density of states of $CoTeMoO_6$.

ACCEPTED

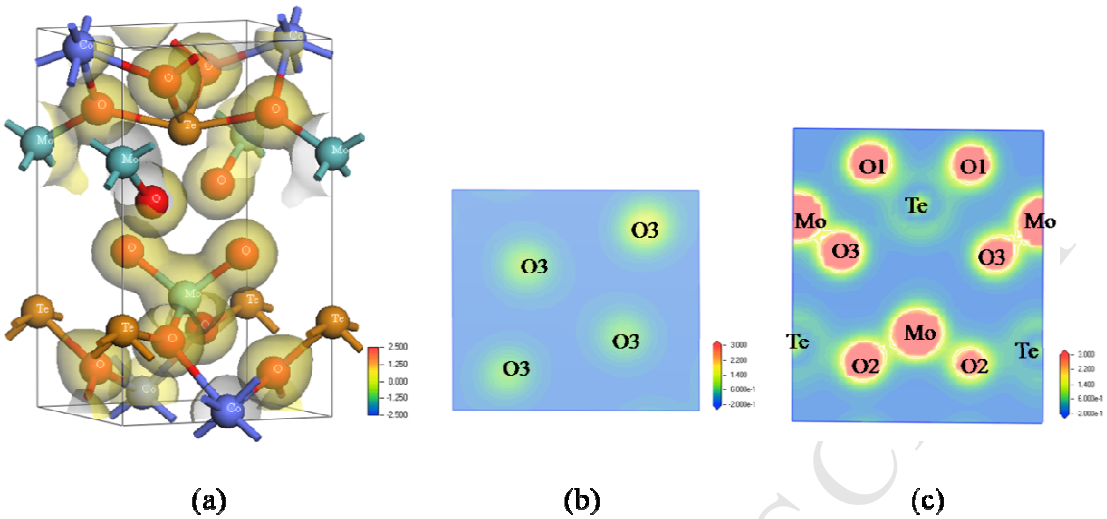

Fig.5 (a) Charge density distributions of $CoTeMoO_6$ crystal structure; (b) total charge density of $CoTeMoO_6$ for a selected (002) plane; (c) total charge density of $CoTeMoO_6$ for a selected (110) plane.

ACCEPTED MANUSCRIPT

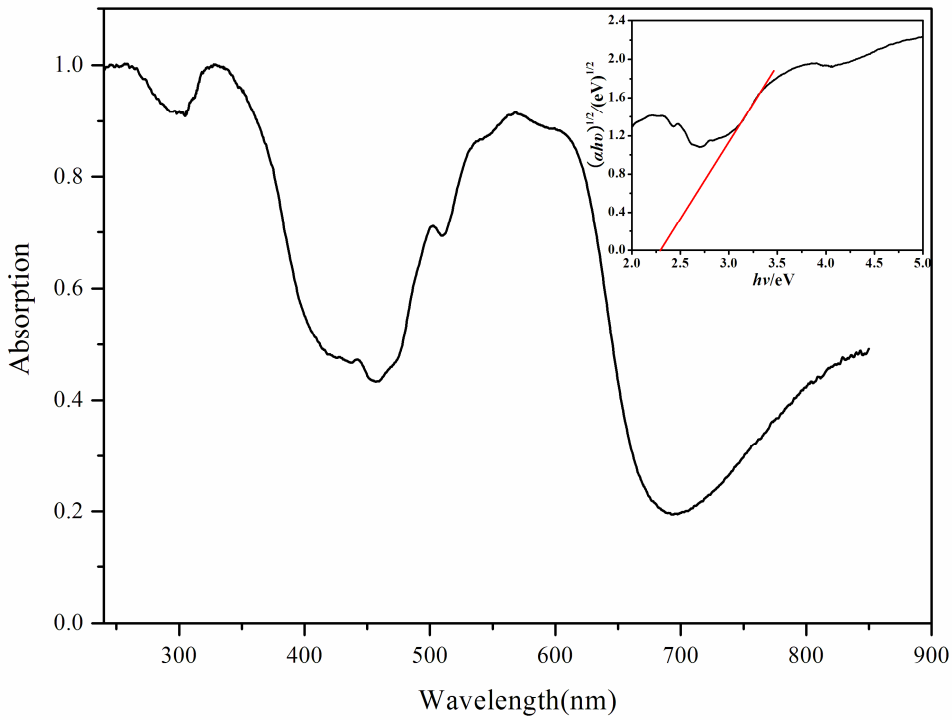

Fig.6 UV-vis diffuse reflectance spectroscopy plots for $CoTeMoO_6$. The top right inset shows the plots of $(\alpha h\nu)^{1/2}$ vs. energy ($h\nu$) for the band gap energies. Note that the band gap is 2.29 eV for $CoTeMoO_6$.

ACCEPTED

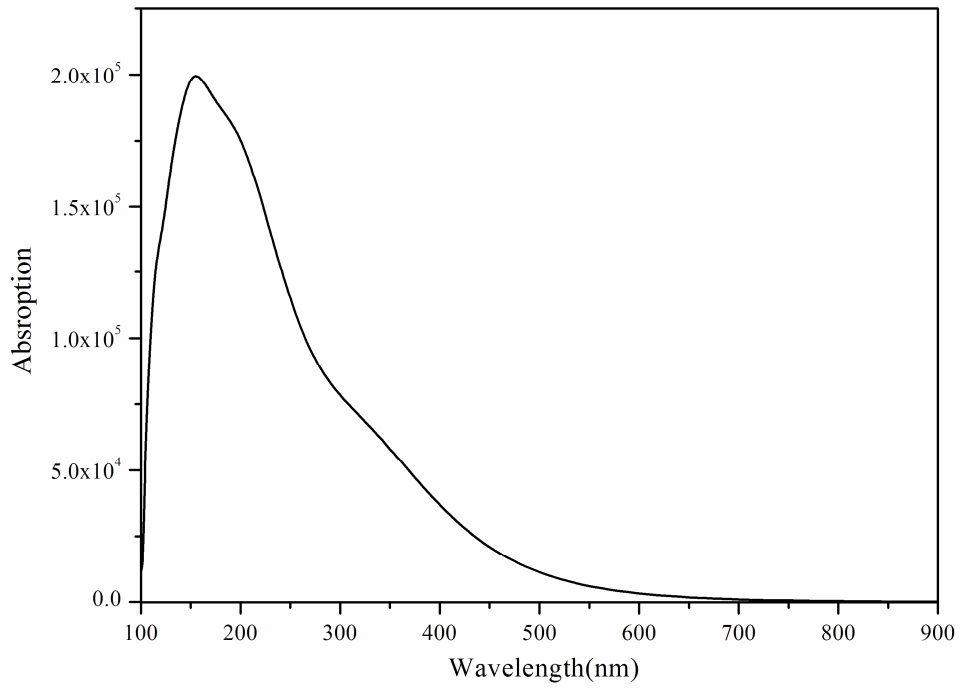

Fig.7 Calculated result of absorption spectra for $CoTeMoO_6$.

ACCEPTED

Supporting information

Table S1 Atomic coordinates and equivalent isotropic displacement parameters (Å^2) for $CoTeMoO_6$ determined using the single-crystal XRD

atom	Wyckoff	occ	*x*	*y*	*z*	U_{eq} (Å^2)
Co(1)	2a	1	0	0.5000	0. 3076(1)	0.007(1)
Te(1)	2c	1	0	1	0. 2468(1)	0.006(1)
Mo(1)	2c	1	0.5000	0.5000	0.6891(2)	0.006(1)
O(1)	4e	1	0. 2216(11)	1.1788(10)	0.1101(7)	0.007(1)
O(2)	4e	1	0. 2398(11)	0. 6831(11)	0. 2059(7)	0.008(1)
O(3)	4e	1	0. 3469(12)	0. 2890(11)	0. 4256(7)	0.011(1)

ACCEPTED MANUSCRIPT

Table S2 Selected bond lengths (Å) and angles (deg.) for $CoTeMoO_6$ [a]

Co(1)-O(1)#3	2.078(5)	Co(1)-O(1)#4	2.078(5)
Co(1)-O(1)#5	2.152(5)	Co(1)-O(1)#2	2.152(5)
Co(1)-O(2)	2.210(6)	Co(1)-O(2)#6	2.210(6)
Te(1)-O(1)#2	1.894(6)	Te(1)-O(1)	1.894(6)
Te(1)-O(2)#2	2.088(6)	Te(1)-O(2)	2.088(6)
Mo(1)-O(3)	1.704(6)	Mo(1)-O(3)#1	1.704(6)
Mo(1)-O(2)#1	1.859(6)	Mo(1)-O(2)	1.859(6)
O(3)-Mo(1)-O(3)#1	104.8(4)	O(3)-Mo(1)-O(2)#1	106.3(3)
O(3)#1-Mo(1)-O(2)#1	107.9(3)	O(3)-Mo(1)-O(2)	107.9(3)
O(3)#1-Mo(1)-O(2)	106.3(3)	O(2)#1-Mo(1)-O(2)	122.4(4)
O(1)#2-Te(1)-O(1)	101.0(4)	O(1)#2-Te(1)-O(2)#2	86.6(2)
O(1)-Te(1)-O(2)#2	80.8(2)	O(1)#2-Te(1)-O(2)	80.8(2)
O(1)-Te(1)-O(2)	86.6(2)	O(2)#2-Te(1)-O(2)	160.2(3)
O(1)#3-Co(1)-O(1)#4	109.0(3)	O(1)#3-Co(1)-O(1)#5	101.43(14)
O(1)#4-Co(1)-O(1)#5	101.41(15)	O(1)#3-Co(1)-O(1)#2	101.41(15)
O(1)#4-Co(1)-O(1)#2	101.43(14)	O(1)#5-Co(1)-O(1)#2	140.1(3)
O(1)#3-Co(1)-O(2)	169.3(2)	O(1)#4-Co(1)-O(2)	81.2(2)
O(1)#5-Co(1)-O(2)	79.1(2)	O(1)#2-Co(1)-O(2)	72.7(2)
O(1)#3-Co(1)-O(2)#6	81.2(2)	O(1)#4-Co(1)-O(2)#6	169.3(2)
O(1)#5-Co(1)-O(2)#6	72.7(2)	O(1)#2-Co(1)-O(2)#6	79.1(2)
O(2)-Co(1)-O(2)#6	88.8(3)		

[a] Symmetry transformations used to generate equivalent atoms: #1 x+1, -y+1, z; #2 -x, -y+2, z; #3 x-1/2, -y+3/2, -z; #4 -x+1/2, y-1/2, -z; #5 x, y-1, z; #6 -x, -y+1, z; #7 -x+1/2, y+1/2, -z; #8 x, y+1, z.

ACCEPTED MANUSCRIPT

Table S3 Anisotropic displacement parameters ($Å^2 \times 10^3$) for $CoTeMoO_6$

atom	U11	U22	U33	U23	U13	U12
Co(1)	6(1)	4(1)	11(1)	0	0	-1(1)
Te(1)	6(1)	3(1)	8(1)	0	0	0(1)
Mo(1)	6(1)	5(1)	9(1)	0	0	1(1)
O(1)	6(2)	3(2)	13(2)	1(2)	5(2)	0(2)
O(2)	7(2)	5(2)	13(2)	1(2)	1(2)	-1(2)
O(3)	10(2)	12(3)	12(3)	-1(2)	1(2)	-2(2)

ACCEPTED MANUSCRIPT

ACCEPTED MANUSCRIPT

Highlights

1. Optical and nonlinear optical properties of $CoTeMoO_6$ were measured.
2. Band structures, DOS, optical properties of $CoTeMoO_6$ were calculated.
3. SHG responses of $CoTeMoO_6$ with 1.06μm laser are much weaker than that of KDP.
4. Absorption band at 458~692 nm has a great effect on SHG response of $CoTeMoO_6$.

ACCEPTED MANUSCRIPT